ICAR - Central Research Institute for Dryland Agriculture

Crop-Water Requirements and Contingency Planning for Semi-Arid Tropics (2

1. Agro-Climatic Zone Context: Marathwada & Western Scarcity Zone

The scarcity zone of Maharashtra receives an average annual rainfall between 500-750 mm. In drought

2. Millets - Water Requirements and Drought Tolerance

Bajra (Pearl Millet) requires 300-400 mm of water throughout its growing period of 75-85 days. It is the

3. Pulse Crops - Rainfed Cultivation & Nitrogen Fixation

Tur (Pigeon Pea / Arhar) has a deep taproot system penetrating up to 2 meters, enabling access to dee

Mung Bean (Green Gram) requires only 300-350 mm of water with a short maturity window of 60-75 days. It acts as an effective catch crop and minimizes economic exposure during erratic rainfall onset. Recommended varieties include BM-4 and Virat.

4. Soil Moisture Conservation & Protective Irrigation

Under severe rainfall deficit (>20% below normal), farmers should apply straw mulching at 5 tonnes per